

Scientific Formalization of Continuous Quantum Fluid Dynamics: The Agora Verification Protocols

SocrateAI Scientific Agora

June 2026

Note (2026-09-26). Several claims below have since been corrected or retracted; see `RETRACTIONS.md` in the repository for the full record. In summary: the “non-linear Quantum Volterra echo” (QVE-02) is not a physical echo (R2); the QV-01 paragraph below describing an RPA-driven “onset of Landau Damping” was never actually implemented that way and has since been replaced by an exact Padé root-finding method with a documented negative result in weak coupling (R6, R7); the Q-RIP-03 manifold is T^2 in the code, not $\mathbb{RP}^1$ as stated below, and $L_* = 4$ is a conjecture/definitional identity, not a proven protective bound (R3, R4); and the Q-RHK-02 “Theorem 22.6” attribution has been checked against the source and found mismatched (R8). The text below is kept as a historical record and is annotated in place rather than rewritten.

Note (2026-09-27): results since this report. The current account is `docs/agora_engine_paper.pdf`. The main points are:

- **QV-01.** $F_0^s = 93/10$ below was a stand-in. On real ${}^3\text{He}$ parameters $F_0^s(P)$ (Kollar & Vollhardt 2000, Table IX, from Greywall 1983) the one-parameter model gives zero sound *slower* than first sound, which is wrong. Adding F_1^s fixes this ($c_0/c_1 = 1.036 \rightarrow 1.005$ from 0 to 29 bar), and the exact Padé solver now handles (F_0^s, F_1^s) . The weak-coupling bracket $2e^{-(2+2/F)} \leq s - 1 \leq F$ is proved in `ZeroSoundBracket.lean`.
- **QVE-02.** The input $\text{sinc}(t)$ is proved to be a free-transport phase-mixing mode that decays like $1/t$ (`PhaseMixingLink.lean`).
- **Q-RHK-02.** Theorem 22.6, applied literally to the kernel below, gives $\bar{\gamma} = 2\sqrt{3m/M} = 3.458$. This number replaces, and must not be compared with, γ_{bound} (R8).
- **Data.** Open ${}^4\text{He}$ neutron dispersion data do exist (Godfrin et al. 2021; R9).

Abstract

This report documents the formalized mathematical physics protocols executed within the SocrateAI Scientific Agora. Strict adherence to the “Zéro Simulation Flottante” policy is maintained throughout: all continuous physical phenomena, including Landau Damping, non-linear Quantum Volterra echoes, and Roton scattering kernels, are extracted algebraically over the rational field $\mathbb{Q}$ avoiding any floating-point hallucinations. The resulting exact sequences are subsequently checked by the Lean 4 kernel utilizing the Mathlib `Rat` type, which certifies the exact rational arithmetic performed on those sequences (see the note above and `RETRACTIONS.md` for what this does and does not establish about the physics). This multi-stage pipeline orchestrates symbolic computation across specialized processing layers to achieve formal verification of that arithmetic.

1 Introduction

The primary objective of the SocrateAI Scientific Agora is the absolute formal verification of theoretical physics bounds. This removes the uncertainties surrounding conventional numerical solvers. We enforce this via a dual-stage workflow: the `LinearResponseStage` handles the exact quantum fluid extraction, while the `KineticStage` executes the theoretical constraint bounding and mapping to the exact rational field $\mathbb{Q}$ for Lean 4 theorem proving.

2 Protocol QV-01: Quantum Vlasov Zero-Sound Extraction

In the high-frequency limits of liquid ^3He , Zero-Sound waves propagate collisionlessly. The exact infinite rational series for the continuous base response (Lindhard function) is given by:

$$\chi_0(z) = \sum_{k=1}^{\infty} \frac{1}{2k+1} z^{2k} \quad (1)$$

where $z = \frac{qv_F}{\omega}$. Introducing the Landau interaction parameter F_0^s , the Random Phase Approximation (RPA) response is:

$$\chi_{RPA}(z) = \frac{\chi_0(z)}{1 - F_0^s \chi_0(z)} \quad (2)$$

Update (see RETRACTIONS.md R6/R7): the RPA-driven pole tracking described above as originally planned was never actually implemented – an early attempt fed a polynomial into a function expecting Taylor coefficients and produced meaningless output, and separately violated the exact-arithmetic rule via a float round-trip. It has been replaced by the following, which is implemented and tested. Instead of floating-point numerical integration, the exact diagonal $[M/M]$ Padé approximant of the Landau zero-sound kernel $\chi(s)$ is computed strictly over $\mathbb{Q}$, and the dispersion relation $\chi(s) = 1/F_0^s$ is solved exactly as a polynomial root. For strong coupling (e.g. $F_0^s = 93/10$, ^3He -like) this gives $s = \sqrt{37/10} \dots$ at $[1/1]$, agreeing with a 60-digit reference to 6.8×10^{-10} at $[4/4]$. For weak coupling, no admissible root exists at any order – the mode is exponentially close to the particle-hole continuum edge, $s - 1 \simeq 2e^{-2-2/F_0^s}$, and this negative result is characterised analytically rather than “tracked into” the continuum, which the method cannot do.

3 Protocol QVE-02: Second-order Volterra response (not an echo – see RETRACTIONS.md R2)

Demonstrating cross-domain Stage-to-Stage integration, the `LinearResponseStage` generates the $\mathcal{O}(\epsilon^1)$ linear density perturbation sequence $\rho^{(1)}(t)$. The `KineticStage` ingests this to compute the induced electric field $E^{(1)}(t)$, subsequently evaluating the exact Cauchy convolution product of the non-linear source term $S^{(2)} = \rho^{(1)} \cdot E^{(1)}$. This exact Volterra integral expansion yields the $\mathcal{O}(\epsilon^2)$ density sequence, which is the Taylor series of $\text{Si}(t)^2/2$ – not a plasma echo, which is a large-time phenomenon requiring two pulses at distinct wavenumbers, neither present here. The computation itself is exact and correct; only the “echo” label is retracted. This prevents discretization drift completely, securing the expansion for Lean 4.

4 Protocol Q-RHK-02: Roton Fractional Heat Kernels

To evaluate the stability of Pitaevskii Plateau phenomena in superfluid ^4He , a phenomenological angular scattering kernel $\beta(\cos \theta)$ is extracted:

$$\beta(\cos \theta) = \frac{1}{2}(1 + \cos^2 \theta) \exp(-0.1(1 - \cos \theta)) \quad (3)$$

We compute the following combination, previously attributed to Villani’s Theorem 22.6 (arXiv:2501.00925) but now checked against the source and found mismatched — Theorem 22.6 is real but bounds a different, multiplicative quantity ($2\sqrt{d} \cdot \sqrt{m_r/M_r}$); the combination below is an independently-defined convention of this project (see RETRACTIONS.md R8):

- Spherical Curvature Integration: $\Sigma(\beta) \approx 1.899$
- Kernel Infimum and Supremum: $m_r \approx 0.4513$, $M_r = 1.0$
- Admissible Singularity Limit: $\gamma_{\text{bound}} \approx 1.9513$

These constants define the critical Fisher Information dissipation bound required to ensure smooth equilibration.

5 Protocol Q-RIP-03: 2D Quantum Ripplons (conjecture stage)

Evaluating topological phase-mixing of capillary waves (ripplons) on a 2D flat topology (T^2 in the implementation – an earlier draft of this document said $\mathbb{RP}^1$, a discrepancy recorded in RETRACTIONS.md R4 and corrected here), the differential Bakry-Émery curvature is conjectured, not derived: with $L_*(d) := 2d$ taken as a definition, $L_*(2) = 4$ follows by arithmetic alone. This is not a proven topological constant, and the claim that it “protects 2D quantum films from Landau damping collapse” has no support here – see RETRACTIONS.md R3 for the full caveat.

6 Conclusion

By leveraging an exact $\mathbb{Q}$ -based extraction layer and deploying subsequent proof resolution using Lean 4 Mathlib tactics (e.g. `norm_num`), the Agora eliminates numerical discrepancies. No simulated pseudo-data is introduced, strictly aligning with empirical rigor.